

Nikhil Ankush Thange

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EDUCATION

Bachelor of Engineering in Information Technology Aug 2023 – Jun 2027 (Expected)
Vasantdada Patil Pratishthan's College of Engineering (Mumbai University) **CGPA: 7.50 / 10.0**
Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, System Design

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL, C/C++, HTML5, CSS3
Frontend: React 19, Next.js, Redux Toolkit, TanStack Query, Tailwind CSS, WebSockets
Backend & Cloud: Node.js, Express, REST APIs, AWS (EC2, S3, DynamoDB, SQS, CloudFront), Docker, Nginx
Databases & Tools: PostgreSQL, MongoDB, Redis, Prisma ORM, Git/GitHub, CI/CD (GitHub Actions), Jest, Vitest
AI & Machine Learning: PyTorch, YOLOv8, Computer Vision, LLM Integration (NVIDIA NIM)

WORK EXPERIENCE

Software Development Engineer Intern Jul 2026 – Present
Chitralai / *React 18, TypeScript, Node.js, Express, AWS (EC2, S3, DynamoDB, SQS), Redis, Docker* Remote

- Frontend Architecture & Landing Page:** Architected a high-converting landing page and attendee portal using **React 18, TypeScript, and Tailwind CSS**; built custom fullscreen media galleries, cutting load times to **under 800ms** and driving a **25% conversion lift**.
- Distributed Backend & Media Pipelines:** Built high-throughput **Node.js/Express** services with direct-to-**S3** presigned multipart uploads; deployed **AWS SQS workers** for media transcoding, eliminating server memory starvation during batch ingestions.
- Cloud Database & Cache Optimization:** Modeled low-latency **AWS DynamoDB** schemas with **Global Secondary Indexes (GSI)**, cutting query latency by **40%**; deployed multi-tier **Redis caching** to stabilize **P95 API latency to sub-65ms**.
- Production Reliability & Root-Cause Debugging:** Diagnosed and resolved critical bottlenecks across S3 storage quotas, EC2 blue-green container routing, and DynamoDB full scans; enforced **85%+ test coverage** in **GitHub Actions CI/CD**.

FEATURED PROJECTS

BMC Smart Civic Operating System (CityOS) [Live Demo](#) | [GitHub](#)
React 19, TypeScript, Node.js, Express, YOLOv8, PyTorch, Redis, MongoDB, Docker, Nginx, Leaflet GIS

- AI Vision Pipeline:** Fine-tuned **YOLOv8** on 6 civic issue classes achieving **91.4% mAP@50**; implemented client-side Laplacian blur rejection and automated feedback loops for continual model training.
- High-Concurrency Backend:** Sustained **500+ concurrent requests at sub-50ms P95 latency** using a Node.js PM2 cluster, connection pooling, and multi-layer Redis TTL caching.
- Geospatial Deduplication:** Integrated MongoDB 2dsphere spatial indexing and Turf.js point-in-polygon algorithms across 24 municipal wards to eliminate duplicate complaints within a **35m radius**.

HireMate – AI Recruitment Intelligence Platform [Live Demo](#) | [GitHub](#)
React 19, Node.js, Express, NVIDIA NIM LLMs, WebSockets, PostgreSQL, MongoDB, JWT

- Low-Latency LLM Streaming:** Built real-time conversational interview simulations by streaming **NVIDIA NIM LLM** tokens over WebSockets, maintaining end-to-end response latency under **300ms**.
- Resume Matching Engine:** Engineered an automated ATS scoring engine using structured prompt validation, reaching **88% semantic alignment accuracy** between candidate resumes and job profiles.
- Enterprise Security:** Implemented role-based access control (RBAC) with stateless JWT authentication, cryptographic password hashing, and rate-limiting middleware.

SwasthyaSetu – Rural Healthcare Continuity Layer [Live Demo](#) | [GitHub](#)
React 19, TypeScript, Node.js, Express, PostgreSQL, Prisma, Dexie.js (IndexedDB), Hugging Face TrOCR, GSM SMS

- Offline-First Architecture:** Architected a client-side **IndexedDB (Dexie.js)** storage queue with idempotent background sync to PostgreSQL/Prisma, ensuring **0% data loss** during network outages.
- Document OCR Integration:** Integrated a **Hugging Face TrOCR** transformer model for automated digitization of handwritten clinical prescriptions.
- Low-Bandwidth Fallback:** Created an emergency referral protocol using a **≤ 160-character GSM SMS** transport payload for healthcare workers in zero-internet rural areas.

HONORS & CERTIFICATIONS

- MUSA Codex National Hackathon:** Top 15 Finalist (2026) – Built SwasthyaSetu offline emergency triage platform.
- IBM Full Stack Software Developer Certification:** Microservices, Cloud Architecture, and RESTful APIs.
- Data Structures & Algorithms:** Solved 250+ coding challenges across LeetCode and GeeksforGeeks.
- Quasar 4.0 Hackathon (2026)** – Engineered a Student Placement portal to automate campus recruitment workflows.